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Solutions for TCS Xplore iPA held on 14-April-26

Java – First Problem Statement

Question

Create a program to count the number of words present in a given string. Create class **Solution** with main method.

Implement the below static method in Solution class.

CountWords(String str): The method should take string as an input and return the total number of words in the string. If the input is null/empty, return 0.

Constraints:

- Words are separated by one or more spaces.
- Leading and trailing whitespaces should be ignored.
- Empty string should return 0.

Sample Input 1:

“Hello World”

Sample Output 1:

2

Sample Input 2:

“ Java is fun”

Sample Output 2:

3

Solution

```
import java.util.*;

public class Solution {

    public static int countWords(String str) {

        if (str == null) {

            return 0;

        }

        str = str.trim();

        if (str.length() == 0) {

            return 0;

        }

        String[] words = str.split("\\s+");

        return words.length;

    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String input = sc.nextLine();

        int result = countWords(input);

        System.out.println(result);

    }

}
```

Sample TestCases

Sample Input 1:

“Hello World”

Sample Output 1:

2

Sample Input 2:

“ Java is fun”

Sample Output 2:

3

Sample Input 3:

“Java is fun”

Sample Output 3:

3

Sample Input 4:

“”

Sample Output 4:

0

Sample Input 5:

“ ”

Sample Output 5:

0

Java – Second Problem Statement

Question

Build a console application to manage the driver and truck related data. Each truck will have exactly one driver who will drive the truck.

Create a class **Truck** with following attribute

id: integer

name: String

totalMilesTraveledByTruck: Float

Create a class **Driver** with following attribute

id: integer

name: String

contact: String

experience: Float

Establish the relationship between the Truck and Driver. Create getters, setters, and parameterized constructors for the above classes.

Create class Solution with the main method.

Implement the below static method in Solution class

getAllTruckWithDistance method:

This method accepts an array of Truck objects and an integer value representing the threshold distance as input parameters. It returns an array of Truck objects whose total miles traveled is greater than or equal to the specified distance.

The resulting array must be sorted in descending order based on the associated driver's experience.

If no truck meets the specified distance criteria, the method should return null.

Note:

- A Truck must have exactly one associated Driver, and a driver cannot exist without a Truck.
- If multiple drivers have the same experience value, the corresponding trucks should be ordered according to the original insertion order in the input array.

Test Cases:

Sample input 1:

4

101

EcoHaul

2999.99

11

Arjun Nair

9876500101

3.0

102

CargoJet

3000

12

Priya S

9876500102

7.5

103

HighwayStar

4500.5

13

Rohit Menon

9876500103

5.25

104

MetroLink

3000.00

14

Nikita Rao

9876500104

9.0

3000

Sample output 1:

104

MetroLink

3000.0

14

Nikita Rao

102

CargoJet

3000.0

12

Priya S

103

HighwayStar

4500.5

13

Rohit Menon

Sample input 2:

0

100

Sample output 2:

No Truck Found

Solution:

```
import java.util.ArrayList;

import java.util.Collections;

import java.util.Comparator;

import java.util.List;

import java.util.Scanner;

public class TruckAndDriverSoltion {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        int size = scanner.nextInt();

        Truck[] trucks = new Truck[size];

        for (int i = 0; i < size; i++) {

            int id = scanner.nextInt();

            scanner.nextLine();

            String name = scanner.nextLine();

            float totalMiles = scanner.nextFloat();

            int did = scanner.nextInt();

            scanner.nextLine();

            String dname = scanner.nextLine();

            String dcontact = scanner.nextLine();

            float experiance = scanner.nextFloat();

            Driver driver = new Driver(did, dname, dcontact, experiance);

            trucks[i] = new Truck(id, name, totalMiles, driver);

        }

        //
```

```

//          for (Truck truck : trucks) {
//              System.out.println(truck);
//          }
//

float distance = scanner.nextFloat();

Truck[] output = getAllTruckWithDistance(trucks, distance);

if(output == null) {
    System.out.println("No Truck Found");
}else {
    for(Truck truck : output) {
        System.out.println(truck.getId());
        System.out.println(truck.getName());
        System.out.println(truck.getTotalMilesTraveled());
        System.out.println(truck.getDriver().getId());
        System.out.println(truck.getDriver().getName());
    }
}

scanner.close();
}

private static Truck[] getAllTruckWithDistance(Truck[] trucks, float distance) {
    List<Truck> input = new ArrayList<Truck>();

    if(trucks.length == 0) {

```

```
return null;
```

```
}
```

```
for (Truck truck : trucks) {
```

```
if (truck.getTotalMilesTraveled() >= distance) {
```

```
input.add(truck);
```

```
}
```

```
}
```

```
Collections.sort(input, Comparator.comparing(obj -> ((Truck)  
obj).getDriver().getExperience()).reversed());
```

```
Truck[] output = input.toArray(new Truck[input.size()]);
```

```
if(output.length == 0) {
```

```
return null;
```

```
}
```

```
return output;
```

```
}
```

```
}
```

```
class Driver {
```

```
private int id;
```

```
private String name;
```

```
private String contact;
```

```
private float experience;
```

```
public Driver() {  
    super();  
}  
  
public Driver(int id, String name, String contact, float experience) {  
    super();  
    this.id = id;  
    this.name = name;  
    this.contact = contact;  
    this.experience = experience;  
}  
  
public int getId() {  
    return id;  
}  
  
public void setId(int id) {  
    this.id = id;  
}  
  
public String getName() {  
    return name;  
}  
  
public void setName(String name) {  
    this.name = name;  
}  
  
public String getContact() {  
    return contact;  
}
```

```

public void setContact(String contact) {
    this.contact = contact;
}

public float getExperience() {
    return experience;
}

public void setExperience(float experience) {
    this.experience = experience;
}

@Override
public String toString() {
    return "Driver [id=" + id + ", name=" + name + ", contact=" + contact + ", experience=" +
        experience + "]";
}
}

class Truck {
    private int id;
    private String name;
    private float totalMilesTraveled;
    private Driver driver;
    public Truck() {
        super();
    }
    public Truck(int id, String name, float totalMilesTraveled, Driver driver) {
        super();
    }
}

```

```
this.id = id;

this.name = name;

this.totalMilesTraveled = totalMilesTraveled;

this.driver = driver;
}

public int getId() {
    return id;
}

public void setId(int id) {
    this.id = id;
}

public String getName() {
    return name;
}

public void setName(String name) {
    this.name = name;
}

public float getTotalMilesTraveled() {
    return totalMilesTraveled;
}

public void setTotalMilesTraveled(float totalMilesTraveled) {
    this.totalMilesTraveled = totalMilesTraveled;
}

public Driver getDriver() {
    return driver;
}
```

```
}  
  
public void setDriver(Driver driver) {  
  
    this.driver = driver;  
  
}  
  
@Override  
  
public String toString() {  
  
    return "Truck [id=" + id + ", name=" + name + ", totalMilesTraveled=" + totalMilesTraveled +  
    ", driver=" + driver + "];"  
  
}  
  
}
```

Test Case

Sample input 1:

4

101

EcoHaul

2999.99

11

Arjun Nair

9876500101

3.0

102

CargoJet

3000

12

Priya S

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103

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13

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9876500103

5.25

104

MetroLink

3000.00

14

Nikita Rao

9876500104

9.0

3000

Sample output 1:

104

MetroLink

3000.0

14

Nikita Rao

102

CargoJet

3000.0

12

Priya S

103

HighwayStar

4500.5

13

Rohit Menon

Sample input 2:

0

100

Sample output 2:

No Truck Found

Sample input 3:

3

201

IronTrail

1800

21

Vivek Gupta

022-6001001

4.0

202

BlueComet

1800.00

22

Sneha Kulkarni

022-6001002

4.0

203

PolarVoyage

5000.75

23

Manish Agarwal

022-6001003

6.0

0

Sample output 3:

203

PolarVoyage

5000.75

23

Manish Agarwal

201

IronTrail

1800.0

21

Vivek Gupta

202

BlueComet

1800.0

22

Sneha Kulkarni

Sample input 4:

3

301

NightRunner

-5

31

Sara Khan

skhan@example.com

2.0

302

DesertStar

1999.99

32

Vikram Rao

9999912345

-1.0

303

CityLine

0

33

Leena D

leena_d

0.0

2500

Sample output 4:

No Truck Found

=====

Sample input 5:

4

801

Blue Horizon

4500

81

Ananya Sen

Tel 033 4000 1001

2.75

802

Thunder Road

3000.50

82

Rohit A

rohit.a@fleet.co

4.25

803

Green Valley

3000

83

Simran Kaur

+1 415 555 0199

6.5

804

Crimson Bay

6500.25

84

Arvind N

ARVIND-OPS-009

9.0

3000

Sample output 5:

804

Crimson Bay

6500.25

84

Arvind N

803

Green Valley

3000.0

83

Simran Kaur

802

Thunder Road

3000.5

82

Rohit A

801

Blue Horizon

4500.0

81

Ananya Sen